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## Femtosecond infrared spectroscopy reveals the primary events of the ferrioxalate actinometer

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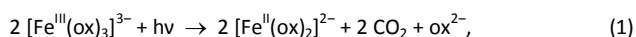
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Chemical actinometry is an indispensable analytical tool in preparative photochemistry that allows for a precise measurement of radiant fluxes inside photoreactors. An actinometer thus enables an absolute determination of the quantum yield of a photochemical reaction of interest. The “gold standard” of chemical actinometry in liquid systems is the Hatchard-Parker actinometer, i.e. an aqueous solution of potassium trisoxalatoferate(III), which is based on the light-induced net transformation of ferric into ferrous oxalate complexes. Although the absolute photochemical quantum yield for this fundamental standard system has been accurately known for many years, the underlying molecular-level mechanisms and time scales associated with a photoreduction of the ferrioxalate actinometer remained so far largely obscured. Here, we use femtosecond mid-infrared spectroscopy combined with ultrafast laser photolysis to obtain unique structural-dynamical information associated with the primary light-triggered processes thereby finally providing the missing quantitative molecular-level foundations that ultimately justify a utilization of aqueous ferrioxalate as a true photochemical standard. Following photon absorption by the octahedral parent complex, an ultrafast decarboxylation occurs within 500 fs, which generates a penta-coordinated ferrous dioxalate that carries a bent carbon dioxide radical anion ligand in an “end-on” O-coordinated fashion. This unique intermediate structurally isomerizes on a tens of picosecond time scale and subsequently loses a  $\text{CO}_2^{\bullet-}$ -ligand to form a square-planar bisoxalatoferate(II) on a hundreds of picoseconds time scale.

### 1 Introduction

An elucidation of the light-triggered processes of transition metal compounds featuring dicarboxylate ligands is of paramount importance for an understanding of a variety of chemical phenomena in nature and technology. For example, dicarboxylate species of ferric iron are believed to be involved in the photoinduced abiotic degradation of organic matter in natural aquatic bodies where they can serve as a source for highly reactive oxygen species through Fenton/photo-Fenton-type redox chemistry.<sup>1–3</sup> Moreover, advanced oxidation processes relying on  $[\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3]^{3-}$  (i.e. trisoxalatoferate(III) or simply, ferrioxalate) as a photoactive catalyst in the presence of hydrogen peroxide have been tested for wastewater treatment in laboratory settings as well as in pilot plants employing direct solar radiation<sup>4–5</sup> and may thus impact future water remediation technologies.<sup>6</sup> Most importantly, however, the light-induced reactions of  $[\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3]^{3-}$  in water constitute the molecular basis of the “gold standard” of photochemical actinometry, the ferrioxalate actinometer originally introduced by Hatchard and Parker.<sup>7–8</sup> As such, this system is frequently utilized as a precise tool for deter-

mining the photon flux of a photochemical reactor's light source<sup>9</sup> and rests on the effective radiative conversion of iron(III) to iron(II). This photoreduction may conveniently be expressed by the net reaction



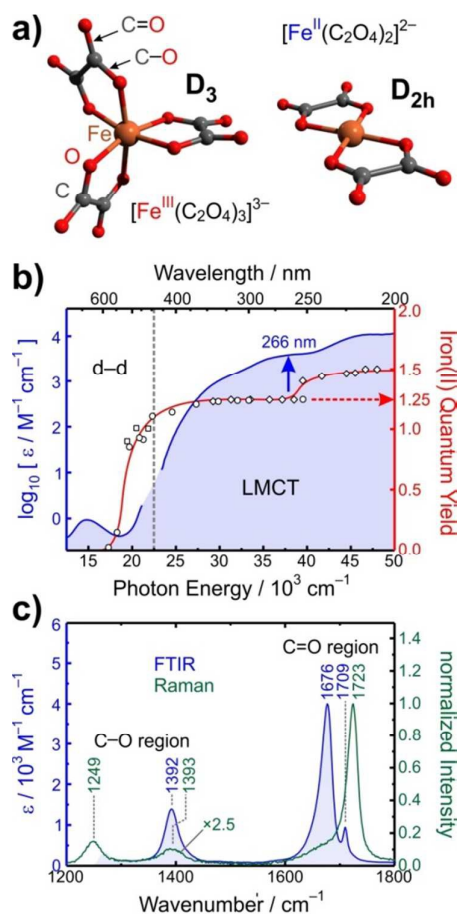
where  $\text{ox} = \text{C}_2\text{O}_4$  (see Fig. 1a for the molecular structures of the reactant and product complexes). The quantum yield for the actinometer conversion in liquid water as a function of the energy of the incident photons is accurately known for more than sixty years and has been critically assessed and refined using various techniques ever since (cf. Fig. 1b, symbols).<sup>8, 10–15</sup>

Paradoxically, however, and despite the recommendation by the International Union of Pure and Applied Chemistry (IUPAC) to use aqueous ferrioxalate solutions in particular for liquid phase actinometry,<sup>7</sup> the molecular-level mechanisms associated with the photochemistry of  $[\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3]^{3-}$  are still entirely unresolved (see supplementary information, SI, for a brief discussion of the various contradicting mechanisms that have been proposed in the literature so far).<sup>10, 16–28</sup> In brief, the various mechanisms essentially differ with respect to the temporal ordering of the one-electron reduction of the iron center (from oxidation state +III to oxidation state +II) on the one hand and the bond breakages within the ligand sphere that are required for a  $\text{CO}_2$ -release on the other (i.e. “prompt” versus “delayed” electron transfer).

In light of the highly diverging views on the primary processes of aqueous ferrioxalate, some of which were recently derived from

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**Fig. 1.** Structure and linear spectroscopy of ferrioxalate. (a) DFT-optimized structures (see text for more details), labeling of the oxalate's stretching coordinates, and symmetries of the ferric (parent) and ferrous (product) oxalato complexes of the actinometer reaction (1). (b) UV/Vis spectrum of ferrioxalate in liquid  $D_2O$  and net photochemical quantum yields for  $Fe(II)$  formation from Refs. [8, 10, 14]. (c) FTIR-spectrum (blue) and Raman-spectrum (green) of aqueous ferrioxalate in the oxalate's C=O and C-O stretching regions. Note that the Raman-spectrum has been scaled in the C-O region by a factor of 2.5 for better visibility

state-of-the-art ultrafast X-ray absorption spectroscopy,<sup>21-23, 27-29</sup> and in light of the fundamental importance of the ferrioxalate actinometer for the general field of photochemistry, experiments are urgently required, which provide unambiguous structural and dynamical information related to the evolution of the metal's coordination sphere and the nature of the emerging fragments following the initial event of photon absorption. Here, we present femto-second (fs) UV-pump/mid-infrared-probe (UV/ MIR) spectroscopy to provide exactly such missing data. We employed UV-pump pulses (center wavelength of 266 nm) to excite the ferrioxalate anions well into their ligand-to-metal charge-transfer region (cf. Fig. 1b for their electronic absorption spectrum in aqueous solution) as was done in the previous ultrafast x-ray studies.<sup>27</sup> Infrared (IR) pulses tunable between  $1000\text{ cm}^{-1}$  and  $3600\text{ cm}^{-1}$  were used for probing the ensuing photo-induced dynamics through the vibrational resonances of the ferrioxalate parent complex and those of the nascent photolysis products (see SI for experimental details).

## 2 Results and Discussion

### 2.1 Linear Spectroscopy

The linear UV/Vis absorption spectrum of aqueous ferrioxalate is displayed in Fig. 1b. It can formally be divided into two spectral regions: the visible part for wavelengths longer than 450 nm, which is governed by the weak ligand-field (d-d) transitions, and the ultra-violet part ( $\lambda \leq 450\text{ nm}$ ), which features a series of very intense and strongly overlapping ligand-to-metal charge transfer transitions (LMCT, oxalate-to-iron). Fig. 1b also displays experimentally determined photochemical quantum yields,  $\Phi$ , for the actinometer reaction (1). The steep rise of  $\Phi$  near  $\lambda \approx 500\text{ nm}$  must be attributed to the onset of LMCT excitations. It can be seen that deeply in the UV, the quantum yield clearly exceeds a value of 1 thereby signaling that one photon is able to convert more than one ferric parent complex and that the net actinometer transformation must be composed of a sequence of multiple elementary reactions. Here, we used a photolysis wavelength of 266 nm to ascertain that only LMCT states are excited (see vertical arrow in Fig. 1b).

The linear (i.e. stationary) Fourier-transform infrared (FTIR) spectrum of ferrioxalate in heavy water ( $D_2O$ ) solution at room temperature is shown in Fig. 1c. The molecular structure of the complex belongs to the point group  $D_3$  and as such exhibits helical chiral symmetry. In this work, however, a racemic mixture of the  $\Delta$  and  $\Lambda$  enantiomers was used for simplicity. In  $D_3$ -symmetry, the vibrational modes of  $[Fe(III)(C_2O_4)_3]^{3-}$  can be classified into totally symmetric modes of species  $A_1$ , which are IR forbidden but Raman allowed, antisymmetric modes of species  $A_2$ , which are IR allowed and Raman forbidden, and finally, doubly degenerate asymmetric modes of species E, which are both IR and Raman allowed.<sup>30-31</sup> All bands seen in Fig. 1c are due to vibrations that are localized exclusively on the three oxalate ligands. The resonance peaking at  $1392\text{ cm}^{-1}$  is mostly caused by the E-modes involving the stretching motion of the oxalates' inner C-O bonds that are directly connected to the central Fe-center. From here on, this band will therefore be labeled E(C-O). The principal band peaking at  $1676\text{ cm}^{-1}$  originates mostly from the E-modes associated with a stretching motion of the oxalate's peripheral C=O bonds. It will therefore be labeled E(C=O). Finally, an additional minor side band can be seen at  $1709\text{ cm}^{-1}$ , whose origin becomes clear upon inspection of the complementary Raman spectrum of ferrioxalate (cf. Fig. 1c, green curve).

In the same wavenumber region, the Raman spectrum is dominated by a single peak at  $1723\text{ cm}^{-1}$  that features a pronounced low-frequency tail extending out to more than  $1650\text{ cm}^{-1}$ . This Raman-band must be assigned to the totally symmetric C=O stretch ( $A_1$ ), which is strictly IR-forbidden in  $D_3$ -symmetry. However, its spectral overlap with the minor IR-band at  $1709\text{ cm}^{-1}$  presents evidence that in aqueous solution the complex experiences static and/or dynamic symmetry breaking, which gives rise to IR-activity of the otherwise IR-forbidden  $A_1$ -modes and to Raman-activity of the normally Raman-silent  $A_2$ -modes. Note however that the ferrioxalate trianion contains a  $d^5$ -configured high-spin ( $S = 5/2$ , sextet) iron(III) center in an pseudo-octahedral ligand sphere and therefore, a static Jahn-Teller distortion cannot be held accountable for symmetry breaking. Moreover, the Raman activity of the  $A_1$ -symmetrical C=O-mode is largest when the complex has perfect  $D_3$ -

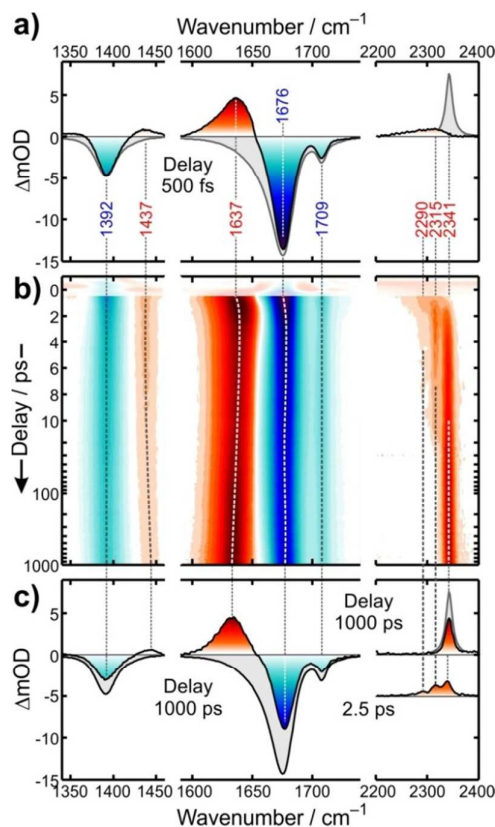
symmetry, i.e. when its IR-cross section vanishes exactly. This rationalizes why the minor IR-side band at  $1709\text{ cm}^{-1}$  does not line up exactly with the major Raman-band at  $1723\text{ cm}^{-1}$ . From here on, the IR peak at  $1709\text{ cm}^{-1}$  will thus be labeled  $A_1(\text{C}=\text{O})$ .

## 2.2 Pump-Probe Spectra

A fs-UV-pump/MIR-probe-spectrum of ferrioxalate in heavy water at room temperature is shown in Fig. 2a. Such a time-resolved spectrum reports the differential optical density,  $\Delta\text{OD}$ , of the sample, which is equal to the difference of the sample's absorbance in the presence and in the absence of the pump pulse, as a function of the probe wavenumber. Note that the spectrum was recorded at a very short pump-probe time-delay of only 500 fs. In the spectral region of the peripheral and inner CO stretching modes of the oxalate ligands (left and middle wavenumber axes, respectively), the pump-probe data feature distinct signals with negative  $\Delta\text{OD}$  (shaded light to dark blue). These signals spectrally coincide exactly with the  $\text{E}(\text{C}=\text{O})$ ,  $\text{E}(\text{C}=\text{O})$ , and  $\text{A}_1(\text{C}=\text{O})$  bands seen in the FTIR spectrum; namely, at  $1392\text{ cm}^{-1}$ ,  $1676\text{ cm}^{-1}$ , and  $1709\text{ cm}^{-1}$ . Such negative bands are commonly termed ground-state bleach and are caused directly by the optical excitation of the complex and the associated depletion of population in its ground electronic state.

Simultaneously, signals with positive  $\Delta\text{OD}$  (shaded light to dark red) peaking at  $1437\text{ cm}^{-1}$  and  $1637\text{ cm}^{-1}$  can also be observed in the transient fs-UV/MIR spectrum. These signals are called induced absorptions and indicate the presence of species, which are generated in response to the interaction of the ground-state complex with the photolysis pulse. Either the electronically excited state of the parent complex or nascent primary photochemical products of yet to be determined nature can be held accountable for the appearance of positive pump-probe signals. Intriguingly, when probing in the region of the anti-symmetric stretching mode,  $\nu_3$ , of the carbon dioxide product of the net actinometer reaction, a weak very broad induced absorption can be detected (see Fig. 2a, right wavenumber axis). This signal peaks at  $2315\text{ cm}^{-1}$ , i.e. slightly frequency-downshifted from the reported position<sup>32-33</sup> of the IR-band of the  $\nu_3$  fundamental of  $\text{CO}_2$  in liquid water at  $2341\text{ cm}^{-1}$ , possibly indicating that for time delays as short as 500 fs at least a fraction of photoexcited complexes have already been photoconverted and released a  $\text{CO}_2$  molecule.

The pump-probe spectrum subsequently evolves with increasing delay in a highly complex fashion. The full spectro-temporal evolution is depicted in Fig. 2b in a color-coded contour representation where the color scale is adopted from panel a). Representative kinetic traces for selected probe wavenumbers are reproduced in Fig. 3. Firstly, the amplitudes of all three bleaching bands partially recover on a time scale of a few picoseconds. This initial recovery hints at a very fast internal conversion, which leads to a partial refilling of the parent's electronic ground state. As judged by the magnitude of the bleach at a time delay of 10 ps, a fraction of roughly 25% to 35% of all initially photoexcited complexes suffer from this ultrafast nonradiative decay. Subsequently, all three bleaching bands then continue to recover very slightly on a time scale of ca. 100 ps to give rise to an apparent permanent negative  $\Delta\text{OD}(t \rightarrow \infty)$ . The pump-probe spectrum for a delay of 1 ns is depicted in Fig. 2c. Together with the data from Fig. 2a, this spectrum

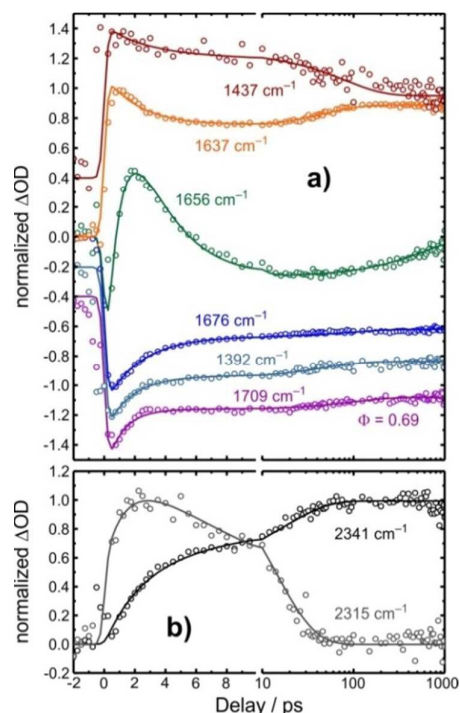


**Fig. 2.** Femtosecond 266-nm-pump/MIR spectroscopy of ferrioxalate in liquid  $\text{D}_2\text{O}$  in the oxalate C–O and C=O stretching regions (left and middle wavenumber axes) as well as in the asymmetric  $\text{CO}_2$  stretching region (right wavenumber axis). (a) Fs-UV/MIR-spectra (induced absorption and bleach are color-coded in red and blue, respectively) recorded at 500 fs after excitation. For comparison, the stationary FTIR spectrum of the sample is also shown in gray (b) Spectro-temporal evolution of the fs-UV/MIR-spectra. Note the logarithmic scaling of the delay axis for delays longer than 10 ps. The dashed curves emphasize the dynamic spectral shifting of the various signal contributions (c) Fs-UV/MIR-spectra recorded after 1 ns.

allows us to quickly estimate a primary photochemical quantum yield,  $\Phi_0$ , from the ratio,  $\Delta\text{OD}(t \rightarrow \infty)/\Delta\text{OD}(t \rightarrow 0)$ , of the final bleaching amplitude to the amplitude initially induced by the UV-pump pulse. Averaging the data from all three bleaching bands yields  $\Phi_0 = 0.65 \pm 0.03$ .

Secondly, the strong induced absorption at  $1637\text{ cm}^{-1}$  initially decays on a time scale of  $\sim 10\text{ ps}$  (cf. Fig. 3a, orange trace) while at the same time, experiencing a distinct dynamic spectral shift to higher probe wavenumbers (cf. white dashed curve in Fig. 2b). Subsequently, this induced absorption displays a pronounced rise on a time scale of about 100 ps, which is accompanied by a spectral back-shift to lower wavenumbers. The weaker induced absorption in the inner C–O region decays also on a 10-ps time scale, but as opposed to the dominant induced absorption from the peripheral C=O region, it continues to decay for delays longer than 10 ps (see Fig. 3a, brown trace). This secondary slower decay is accompanied by a spectral shift to higher probe wavenumbers (cf. gray dashed curve in Fig. 2b).





**Fig. 3.** Time-resolved kinetic traces from femtosecond UV/MIR spectroscopy in the C–O and C=O stretching regions of aqueous ferrioxalate (panel a) and in the antisymmetric CO<sub>2</sub>-stretching region (panel b). Note the logarithmic scaling of the abscissa for delays in excess of 10 ps. The experimental data (symbols) were normalized and vertically shifted for better comparability. The solid curves represent fits to multi-exponential decays/growths.

Thirdly, at a probe wavenumber of 1655 cm<sup>-1</sup>, i.e. in between the strong E(C=O)-bleach and the neighboring strong induced absorption, the dynamics are extraordinarily complex (cf. Fig. 3a, green trace). At early delays, the signal is negative, thus representing an overall bleach. Within 2 ps only, the signal changes sign to become a net induced absorption that slowly converts back into bleach again on a time scale of 10 ps. Finally, this bleach vanishes completely with the observation window of 1 ns. Clearly, this trace demonstrates that during the first nanosecond after LMCT excitation, the ferrioxalate system goes through a complex sequence of competing, parallel, and/or sequential elementary reactions. We have attempted to perform a global analysis of the time- and wave-number-resolved data but the kinetic parameters and species-associated MIR spectra proved uninterpretable because of a large number of singular values, which are required to represent all the dynamically evolving positive and negative signals (vide infra) in a satisfactory manner. The singular value analysis together with decay-associated difference spectra and time-dependent populations of the species is discussed in the supplementary information.

Therefore, only the few selected signals from Fig. 3 were analyzed phenomenologically to document at this stage the most important time scales that can be extracted from kinetics fits to the fs-UV/MIR data. To this end, double-exponential decays and/or growths were used to which a constant offset signal could be added. The corresponding fit parameters are listed in Table 1. Clearly, the fast component with a time constant of less than 2 ps is needed

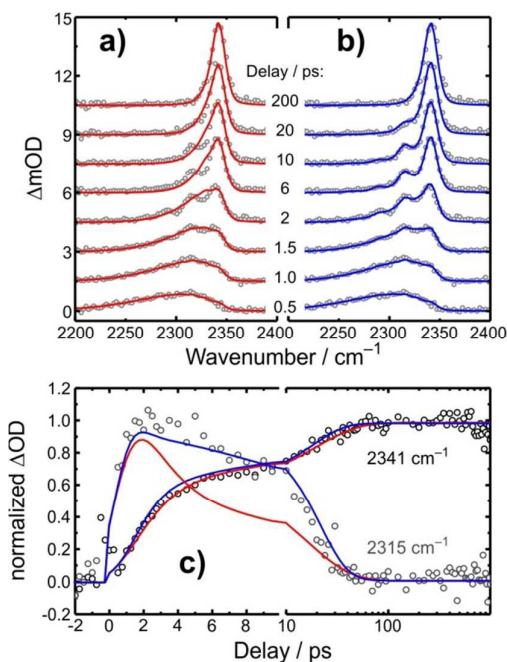
to capture the ultrafast internal conversion process that was identified from the bleaching bands. The time constant around 2.4 ps is required to account for the dynamics of spectral shifting of the induced absorption bands. The time constant of several tens of picoseconds to 100 ps is then needed to reproduce the slower approach of the signals toward their asymptotic limit at 1 ns.

Finally, in the CO<sub>2</sub>-antisymmetric stretching region (Fig. 2, right wavenumber axis), a distinct spectral narrowing and frequency-downshift of the initially very broad absorption is seen within the first few picoseconds. Within this time period, some highly peculiar substructure arises, as exemplified in the inset of Fig. 2c for a time delay of 2.5 ps. After 100 ps, this substructure has disappeared and a single absorption band emerges that peaks at 2341 cm<sup>-1</sup>, i.e. exactly at the spectral position of the ν<sub>3</sub> fundamental transition of aqueous CO<sub>2</sub>. Taken together, these qualitative findings from the CO<sub>2</sub>-stretching region clearly show that the primary carbonaceous fragment is formed within about 100 ps; a delay, after which the spectral shape no longer changes and the time resolved kinetics traces remain constant (cf. Fig. 3b).

**Table 1.** Time constants, τ<sub>i</sub>, and relative amplitudes, A<sub>rel</sub> = A<sub>i</sub> / Σ|A<sub>i</sub>|, of multi-exponential fits to kinetic traces recorded at representative probe wavenumbers.

| ν / cm <sup>-1</sup> | τ <sub>1</sub> / ps (A <sub>rel</sub> ) | τ <sub>2</sub> / ps (A <sub>rel</sub> ) | τ <sub>3</sub> / ps (A <sub>rel</sub> ) |
|----------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| 2341                 | 2.5 (-0.288)                            | 20 (-0.213)                             | ∞ (-0.500)                              |
| 2317                 | 1.5 (-0.370)                            | 14 (0.630)                              | -                                       |
| 1709                 | 1.1 (-0.358)                            | 100 (-0.070)                            | ∞ (-0.572)                              |
| 1676                 | 2.3 (-0.404)                            | 100 (-0.046)                            | ∞ (-0.546)                              |
| 1656                 | 1.1 (-0.538)                            | 2.3 (0.427)                             | 600 (-0.035)                            |
| 1637                 | 2.5 (0.256)                             | 45 (-0.117)                             | ∞ (0.256)                               |
| 1437                 | 2.5 (0.185)                             | 80 (0.278)                              | ∞ (0.537)                               |
| 1391                 | 1.6 (-0.330)                            | 50 (-0.103)                             | ∞ (-0.567)                              |

To be more quantitative, one can estimate the primary quantum yield for CO<sub>2</sub> formation from the signals recorded at the longest time delay of 1 ns. The primary photochemical quantum yield for ferrioxalate conversion was estimated above from the residual bleaching amplitude to be 65%. The extinction coefficient of the A<sub>1</sub>(C=O)-band of ferrioxalate at 1709 cm<sup>-1</sup> is 940 M<sup>-1</sup> cm<sup>-1</sup> (cf. Fig. 1c) while that of the ν<sub>3</sub> fundamental transition of aqueous CO<sub>2</sub> has been reported to be 2300 M<sup>-1</sup> cm<sup>-1</sup>. Thus, if each ferrioxalate complex, which remains bleached after 1 ns, has released exactly one carbon dioxide molecule, the magnitude of the asymptotic induced absorption at 2341 cm<sup>-1</sup> should be a factor of 2300/940 ≈ 2.4 higher than the asymptotic parent bleaching amplitude at 1709 cm<sup>-1</sup>. Inspecting the 1-ns spectrum from Fig. 2c, an absorption-to-bleach ratio of 2.1 is found, i.e. in very good agreement with the expected ratio. Thus, a primary elementary photoreaction occurs with a quantum yield of 65% and releases a single CO<sub>2</sub> from the photoexcited ferrioxalate complex. The remaining 35% of parent complexes undergo internal conversion and return to their electronic ground state structurally intact.



**Fig. 4.** Fs-UV/MIR spectra recorded in the CO<sub>2</sub> antisymmetric stretching region for various representative pump-probe delays. a) complete IVR, b) restricted IVR (see text for details). c) Kinetic traces recorded at the fundamental transition (2341 cm<sup>-1</sup>) and the first hot band (2315 cm<sup>-1</sup>) of the antisymmetric stretching mode. Symbols: experimental data, red curves: complete IVR, blue curves: restricted IVR.

### 2.3 CO<sub>2</sub> vibrational relaxation

The release of a carbon dioxide molecule from the coordination sphere around the iron necessitates the breakage of two chemical bonds; namely, the rupture of one of the two Fe–O bonds between the metal center and the oxalate ligand and the cleavage of the intra-ligand C–C bond. These dynamics may deposit substantial kinetic energy into the internal degrees of freedom of the triatomic fragment and it is quite likely that the peculiar spectro-temporal evolution, which was seen in the CO<sub>2</sub>-stretching region on time scales shorter than 100 ps, is a direct consequence of this internal excitation and the ensuing relaxation of excess vibrational energy.

The data shown in Fig. 3b can again be fitted by double-exponential kinetics. However, the resulting fit parameters (cf. Table 1) describe the dynamics of vibrational relaxation of the CO<sub>2</sub> fragment only in an empirical fashion. To be more accurate, the entire dynamically evolving CO<sub>2</sub> spectra from Fig. 2 have to be modeled in both, the time and the frequency domain on the basis of the full vibrational structure of the carbonaceous fragment. Such a full simulation needs to account for (i) the time-dependent population,  $p_3(t)$  in the vibrational manifold of the antisymmetric stretching mode,  $v_3$ , (ii) the time-dependent populations,  $p_1(t)$  and  $p_2(t)$ , in the manifolds of the symmetric stretching ( $v_1$ ) and bending ( $v_2$ ) degrees of freedom, respectively, (iii) the diagonal anharmonicity,  $x_{33}$ , of the  $v_3$ -mode, and (iv) its off-diagonal anharmonic couplings,  $x_{13}$  and  $x_{23}$ , to the  $v_1$ - and  $v_3$ -modes.

Adopting the model of Hamm et al.<sup>34</sup> and the analysis of Reichardt et al.<sup>35</sup> the infrared band shape of the vibrationally excited linear triatomic CO<sub>2</sub>-molecule can be expressed as

$$S(\tilde{\nu}, t) = \sum_{n_3} [p_3(n_3, t) - p_3(n_3 + 1, t)] \times \sum_{n_1} \sum_{n_2} p_1(n_1, t) p_2(n_2, t) \cdot f(\tilde{\nu} - \tilde{\nu}_a(n_1, n_2, n_3 \rightarrow n_3 + 1)) \quad (2)$$

where  $n_i$  are the vibrational quantum numbers for mode,  $i$ . When a CO<sub>2</sub> molecule is dressed with  $n_1$  quanta in the symmetric stretch and  $n_2$  quanta in the doubly degenerate bend, the anharmonically shifted resonance frequency,  $\tilde{\nu}_3$ , of its anti-symmetric stretching transition from  $n_3$  to  $n_3 + 1$  is then given by

$$\tilde{\nu}_a(n_1, n_2, n_3 \rightarrow n_3 + 1) = \tilde{\nu}_0 + 2x_{33}(n_3 + 1) + x_{13}(n_1 + 1/2) + x_{23}(n_2 + 1/2). \quad (3)$$

Using off-diagonal anharmonic coupling constants from Ref. [36], i.e.  $x_{13} = -19.1$  cm<sup>-1</sup> and  $x_{23} = -12.5$  cm<sup>-1</sup>, in conjunction with a diagonal anharmonicity,  $x_{33} = -13.0$  cm<sup>-1</sup>, a harmonic frequency,  $\tilde{\nu}_0$ , of the  $v_3$ -mode of 2383 cm<sup>-1</sup> is required for the  $v_3$ -fundamental transition to appear at 2341 cm<sup>-1</sup> as observed experimentally.<sup>32–33</sup>

To model the fs-UV/MIR data in the CO<sub>2</sub> stretching region with the above equations, a pseudo-Voigt line shape function

$$f(\tilde{\nu}) = \chi \cdot \exp \left[ -4 \ln 2 \left( \frac{\tilde{\nu}}{\Delta \tilde{\nu}} \right)^2 \right] + (1 - \chi) \cdot \frac{(\Delta \tilde{\nu}/2)^2}{\tilde{\nu}^2 + (\Delta \tilde{\nu}/2)^2} \quad (4)$$

with  $\chi = 0.7$  and an instrument-limited spectral resolution,  $\Delta \tilde{\nu}$ , of 15 cm<sup>-1</sup> (full width at half maximum) was used. Moreover, the populations,  $p_i(n_i, t)$ , were assumed to be Boltzmann distributed

$$p_i(n_i, t) = \frac{g_i(n_i)}{q_i[T_i(t)]} \exp \left( -\frac{E(n_i)}{k_B T_i(t)} \right) \quad (5)$$

where  $k_B$  is Boltzmann's constant,  $T_i(t)$  is the time-dependent vibrational temperature of mode,  $i$ , and  $q_i[T_i(t)]$  its vibrational partition function. Note that the degeneracy of the bending mode is  $g_2(n_2) = n_2 + 1$  while it is equal to 1 for the other two modes.

In the limiting case of ultrafast intramolecular vibrational energy redistribution (complete IVR, much faster than the time required for energy transfer into the surrounding solvent), the internal degrees of freedom of the CO<sub>2</sub> molecule can be regarded as being in thermal equilibrium with each other such that a common canonical temperature,  $T = T_i$ , can be ascribed to all three vibrational modes. Excess energy is dissipated into the surrounding solvent as heat (vibrational energy transfer, VET), thereby leading to a temporal decay of the vibrational temperature without ever disrupting the molecule's internal equilibration at any given time. Here, the temperature is assumed to decay bi-exponentially:

$$\frac{T(t) - T(\infty)}{T(0) - T(\infty)} = \eta \cdot e^{-t/\tau_1} + (1 - \eta) \cdot e^{-t/\tau_2} \quad (6)$$

A comparison between experimental spectra for representative delays and kinetics traces at two different wavenumbers with those obtained from this anharmonic coupling model is displayed in Fig. 4 (symbols and red curves). An optimal fit is obtained with time constants,  $\tau_1 = 1.5$  ps,  $\tau_2 = 20$  ps, a fractional parameter  $\eta = 0.7$ , an initial temperature,  $T(0)$ , of 2400 K and a final temperature,  $T(\infty)$ , of 350 K. A value for  $T(\infty)$  that is slightly higher than room temperature is needed to accurately reproduce the pump-probe spectrum, and in particular, its asymmetric shape at the longest delays. Interestingly, such a small and long-lasting heating of a CO<sub>2</sub>-fragment

has also been observed in the photodissociation of peroxy-carbonates.<sup>35</sup> Note that the 266 nm-photolysis photon deposits a total energy of 450 kJ/mol (37600 cm<sup>-1</sup>) into the system and it could very well be that this excess energy manifests itself as a permanent heating of the pump-probe sample volume within the maximum delay range accessible with our instrument.

More importantly, it can be seen that the time-resolved spectra are reproduced in a satisfactory manner and that the rise of the kinetic trace recorded at the fundamental asymmetric stretching transition,  $|0, 0, 0\rangle \rightarrow |0, 0, 1\rangle$ , at 2341 cm<sup>-1</sup> is modeled very well. However, the same simulation fails to reproduce a complementary trace that is sensitive to the first hot asymmetric stretching transition,  $|0, 0, 1\rangle \rightarrow |0, 0, 2\rangle$ ; namely at 2315 cm<sup>-1</sup>, where the model decays about twice as fast as the experimental data. In addition, the simulated spectra at intermediate delays do not display the distinct substructure mentioned before. It turns out that the fine structure consists of at least three individual sub-bands centered at 2341 cm<sup>-1</sup>, 2315 cm<sup>-1</sup>, and 2290 cm<sup>-1</sup>. The spacing between the lines matches very well the diagonal anharmonicity of the anti-symmetric stretch,  $2x_{33}$ , and therefore, the fine structure can be attributed confidently to the series of  $v_3$ -hot band transitions. Yet, in the simulation, this intriguing progression is washed out by the off-diagonal anharmonic couplings to the other two modes.

This particular discrepancy between experiment and simulation hints at a breakdown of the fast IVR assumption that was made at the outset. Indeed, it may well be that the bending and symmetric stretching vibrations can dissipate their excess energy much faster than the anti-symmetric stretch; a scenario, which requires the process of IVR to be fast but restricted to the bending and symmetric stretching coordinates while at the same time, their equilibration with the antisymmetric stretching mode is slow. This time scale separation essentially decouples the dynamics of VET from the  $v_1$  and  $v_2$ -modes to the solvent from the corresponding dynamics of the  $v_3$ -mode. As a result, the experimental data reflect two parallel and independent cooling processes; namely that of the symmetric stretching and bending modes on the one hand and that of the antisymmetric stretching mode on the other. A similar time scale separation between the relaxation of the antisymmetric stretching mode on the one hand and that of the symmetric stretching and bending modes on the other has also been described by Kandratsenka et al. in the context of molecular dynamics modeling of vibrational cooling of carbon dioxide in non-polar and weakly polar solvents like CCl<sub>4</sub> and CH<sub>2</sub>Cl<sub>2</sub>.<sup>37-38</sup>

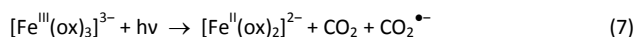
To simulate the IR spectrum of CO<sub>2</sub> in such a restricted IVR scenario, two different vibrational temperatures,  $T_{1,2}(t)$  and  $T_3(t)$ , have to be introduced – one for the  $v_1$  and  $v_2$ -modes and another for the  $v_3$ -mode. For simplicity, these two temperatures are chosen to decay in a mono-exponential fashion while the same initial temperature of 2400 K from the previous complete IVR simulation was kept. Optimized time constants of 1.25 ps and 25 ps were obtained for the relaxation of the  $v_1/v_2$ -modes and the  $v_3$ -mode, respectively. A comparison of the restricted IVR model with the experimental data is shown in Fig. 4 (symbols and blue curves). As can be seen, the agreement between experiment and simulation is excellent and has improved dramatically in both; the spectral domain (panel b) and the time domain (panel c), as compared to the complete IVR

model. This improvement is all the more remarkable given that the number of fit parameters of the restricted IVR model is even smaller than that of the complete IVR model.

Most importantly, the simulation also agrees quantitatively with the experimental data on the shortest time scale. This agreement therefore presents an undeniable piece of evidence that the CO<sub>2</sub>-molecule is expelled within 500 fs only. However, even though the vibrational manifold and hence, the IR spectroscopy, of the nascent triatomic product is highly reminiscent of a free and fully solvated CO<sub>2</sub>, it is difficult to envision that it can escape from the solvent shell of the parent within such a short time. We much rather believe that at a delay of 500 fs, the nascent CO<sub>2</sub> is still in relatively close contact with the other metal-containing fragment, which must also emerge from the initial bond breakages. In this notion, one is tempted to connect the faster component of the vibrational cooling dynamics to the recoil dynamics of the geminate fragments. Following up on this idea, it must then be conjectured that the relative translational motion of the two geminate products couples very strongly to the bending and symmetric stretching vibration of the CO<sub>2</sub> fragment but only very weakly to the antisymmetric stretching vibration, simply because in the restricted IVR model, the  $v_1$  and  $v_2$ -modes relax with a time constant of 1.25 ps while the  $v_3$ -mode relaxes within 25 ps. In general, vibrational relaxation is caused by damping of the solute's internal oscillatory motion as a result of external random forces exerted onto the relaxing vibrational coordinates. Why the frictional forces that are induced by the fragment recoil project so poorly onto the anti-symmetric stretching coordinate, but so well onto the symmetric stretching and bending coordinates, will have to be explored in more detail, e.g. within the framework of classical non-equilibrium molecular dynamics simulation as was done previously for understanding mode-specific CO<sub>2</sub> vibrational cooling.<sup>37-38</sup> It could well be that the different relaxation rates simply reflect the intrinsic frequency-dependence of the frictional forces as described by the classical Landau-Teller theory for vibrational energy transfer. However, we believe that the high charge density of the ferrioxalate trianion is transferred into the metal-containing geminate fragment partner and that the resultant highly directional character of the inter-fragment interactions may play a critical role in discriminating between symmetric and antisymmetric modes. Finally, we emphasize that the above analysis neglects any system-specific phenomena such as a Stark effect on the vibrational structure of CO<sub>2</sub> due to the nearby charge of the metal-containing fragment.

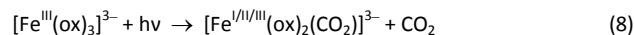
## 2.4 Purely absorptive product spectra

Having verified that with a quantum yield of 65% a single CO<sub>2</sub>-molecule is released from the coordination sphere around the iron center within only 500 fs after photon absorption, we are now confronted with the task to clarify the structure of the metal-containing primary product and its subsequent fate. Formally, there are two distinct pathways that can lead to two non-equivalent Fe-containing primary species: (a) the three-body dissociation, which gives rise to the tetra-coordinated dioxalatoferrate(II) (or equivalently, ferrooxalate) and a carbon dioxide radical anion together with the neutral CO<sub>2</sub> molecule

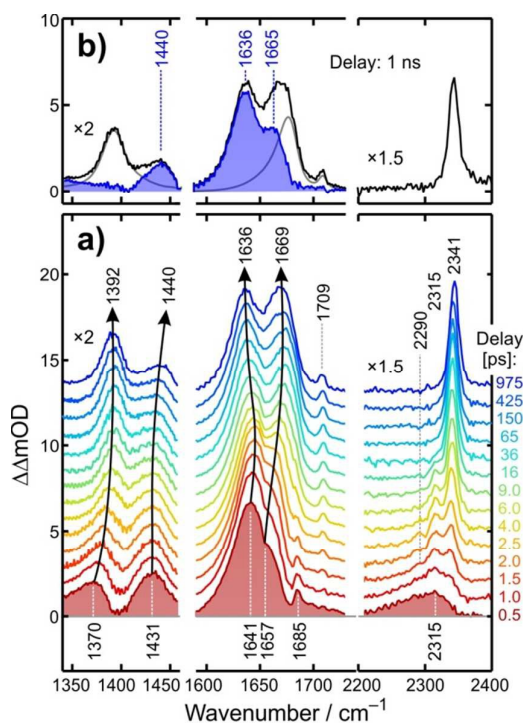




or (b) the two-body dissociation, which yields a penta-coordinated dioxalato-iron primary species that still bears a CO<sub>2</sub>-moiety together with the cleaved CO<sub>2</sub> fragment



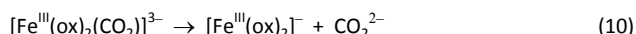
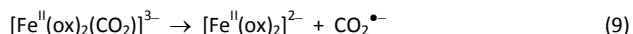
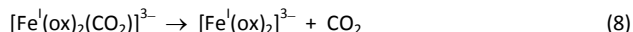
Pathway a) corresponds to a one-electron photoreduction of the Fe-center and generates directly the final ferrous product of the net actinometer reaction (1). The carbonaceous open-shell by-product can continue to react in the bulk with another ferric parent complex thereby rationalizing a net photochemical quantum yield that can be larger than one (cf. Fig. 1b). However, the release of two triatomic fragments from the coordination sphere around the iron would require three simultaneous bond breakages, namely the two Fe–O bonds and the intra-ligand C–C-bond. Considering that the ferrous dioxalate is a planar molecule and the parent has a distorted octahedral ligand sphere (cf. Fig. 1a), a significant structural rearrangement of the two remaining intact oxalates will have to follow or accompany the three-body fragmentation. It would be highly astonishing if all these processes can take place in a time period as short as 500 fs.



**Fig. 5.** a) Purely absorptive product spectra for representative time delays obtained by subtracting the ground state bleaching component from the raw fs-UV/MIR data from Fig. 2. The data from the inner C–O stretching region (left abscissa) have been scaled by a factor of 2. The data from the CO<sub>2</sub> region are also shown for comparison (right abscissa) and have been scaled by a factor of 1.5 for better comparability. The curly arrows serve to highlight the spectral shifting dynamics of the various absorption bands. b) Final product spectrum (black) after 1 ns and the scaled FTIR spectrum of the equilibrated parent (gray). The blue-shaded spectrum arises from photochemical products.

Pathway b) appears at a first glance to be a two-electron photoreduction that generates an iron(II) center to which a neutral CO<sub>2</sub>-ligand is attached. However, it could also represent a redox-neutral

photoreaction that preserves the oxidation state +III at the metal provided the CO<sub>2</sub>-ligand is bound as a dianion (i.e. CO<sub>2</sub><sup>2-</sup>). Moreover, if the CO<sub>2</sub>-moiety is coordinated as a radical mono-anion (i.e. CO<sub>2</sub><sup>•-</sup>) then the oxidation state at the metal may be +II, in which case path b) is also a one-electron photo-reduction like pathway a). At any rate, the penta-coordinated species can be expected to be prone to further decomposition according to



The resultant decay products may attack another ferric parent in the bulk to ultimately generate more than one ferrous dioxalate product thereby explaining again a quantum yield that exceeds a value of one. Moreover, such a two-body fragmentation of a pseudo-octahedral complex may lead to a square-planar (*sqp*) or trigonal-bipyramidal (*tbp*) ligand sphere around the metal center, which in turn may require structural reorganizations of the remaining five ligands that are much smaller than those associated with the three-body dissociation (*vide infra*).

In the regions of the oxalate C=O and C–O-stretching modes, distinct induced absorption signals at 1437 cm<sup>-1</sup> and 1637 cm<sup>-1</sup> were detected in the pump-probe spectra at a delay time of 500 fs (cf. Fig. 2). The amplitudes and spectral positions of these signals subsequently evolve with increasing delay (cf. also Fig. 3). To facilitate an assignment of these induced absorptions, it is highly advantageous to remove in a first step the perturbing bleaching bands from the raw fs-UV/MIR spectral data. This is readily accomplished by adding to the data a properly weighted stationary FTIR spectrum of the sample. To this end, the FTIR data were scaled such that the bleaching band associated with the (forbidden) A<sub>1</sub>-mode at 1709 cm<sup>-1</sup> subtracts smoothly and only a slowly varying background is left behind. The resultant purely absorptive product spectrum<sup>39-40</sup> is depicted in Fig. 5a for a variety of different pump probe time delays along with the corresponding fs-UV/MIR data from the CO<sub>2</sub>-region for comparison. The full spectro-temporal evolution of the purely absorptive product spectrum is shown in a color-coded contour representation similar to Figure 2b in the supplementary information. Note here that the term “product spectrum” maybe misleading at first glance because the resultant absorption data will not only contain signals from real, chemically distinct product molecules but also contributions from the partially replenished electronic ground state of the intact parent complex if its spectrum differs from that of the equilibrated ground state, i.e. the sample’s FTIR spectrum.

In the region of the oxalates’ inner-C–O stretching modes (left wavenumber axis), this spectrum is initially comprised of two well separated bands peaking at 1370 cm<sup>-1</sup> and 1431 cm<sup>-1</sup>, i.e. located on both the low and the high-frequency edges of the parent’s E(C–O)-band at 1392 cm<sup>-1</sup> (cf. Fig. 5a, red-shaded spectrum at the bottom). The component at higher wavenumber was already identified in the raw fs-UV/MIR data but the component at lower wavenumber was mostly obscured by the ground state bleach.

In the region of the peripheral-C=O stretching modes (middle wavenumber axis in Fig. 5a, bottom), an intense broad band peak-



ing at  $1641\text{ cm}^{-1}$  is detected together with a smaller band at  $1685\text{ cm}^{-1}$ , whose relative intensity is comparable to the forbidden  $A_1(\text{C}=\text{O})$ -band of the parent's ground state. Moreover, an additional shoulder riding on the high frequency edge of the  $1641\text{ cm}^{-1}$ -band is discernible from a pronounced saddle point at  $1657\text{ cm}^{-1}$ . Qualitatively, and apart from the shoulder, the overall shape of the early-time product spectrum in the peripheral  $\text{C}=\text{O}$  region appears thus very similar to that of the stationary FTIR spectrum except for an overall frequency-downshift of about  $30\text{ cm}^{-1}$ . At first glance, it seems as if the impulsive electronic excitation transfers the broken symmetry of the parent's ground state fully onto the excited state and into the emerging nascent fragments.

Furthermore, it can also be seen in Fig. 5 that the  $1709\text{ cm}^{-1}$ -absorption, which was prototypical for the symmetry-broken electronic ground state of the ferrioxalate anion, builds up during the first few picoseconds. On the very same time scale, the predominant absorption band around  $1642\text{ cm}^{-1}$  with its distinctive shoulder at  $1657\text{ cm}^{-1}$  gradually splits into two well separated bands, which then settle for very long delays at  $1636\text{ cm}^{-1}$  and  $1669\text{ cm}^{-1}$ , respectively. Apparently, the initial shoulder smoothly transforms into the stationary  $\text{E}(\text{C}=\text{O})$ -band of the ferrioxalate ground state at  $1676\text{ cm}^{-1}$  (cf. the FTIR spectrum in Fig. 1c). Without a doubt, the two spectral components of the purely absorptive product spectrum, i.e. the minor peak at  $1709\text{ cm}^{-1}$  and the stronger band that upshifts from  $1657\text{ cm}^{-1}$  to higher wavenumbers, must be caused by the 35% fraction of all initially photo-excited complexes that return to the ground state via internal conversion. The overall spectro-temporal evolution with the pronounced time-dependent upshift of the  $\text{E}(\text{C}=\text{O})$ -component is then highly indicative for the electronic ground state of the parent complex undergoing vibrational energy relaxation (VER)<sup>34, 39-41</sup> following its repopulation via the ultrafast internal conversion. The complementary dynamic evolution in the inner  $\text{C}-\text{O}$  stretching region corroborates this interpretation: the band initially peaking at  $1370\text{ cm}^{-1}$  progressively shifts to higher wavenumbers on a time scale of several picoseconds thereby smoothly evolving into the characteristic  $\text{E}(\text{C}-\text{O})$ -band of the relaxed parent ground state at  $1392\text{ cm}^{-1}$  (cf. FTIR spectrum in Fig. 1c).

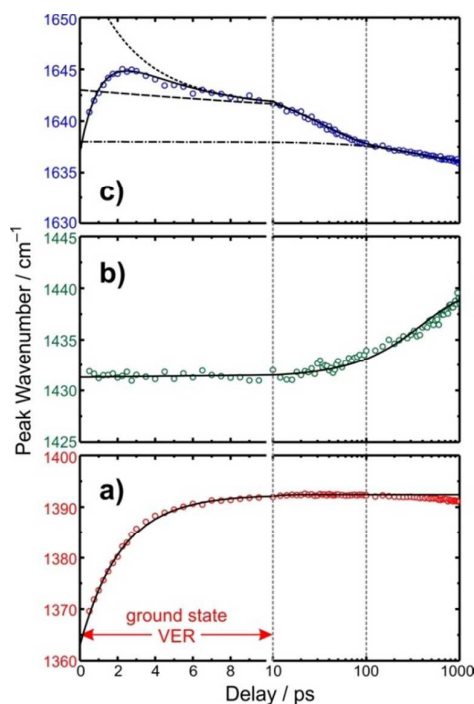
To quantify the ground-state vibrational relaxation dynamics, the spectral peak position of the  $\text{E}(\text{C}-\text{O})$ -band was determined more accurately as a function of the time delay. To this end, a 5th-order polynomial regression was carried out on the purely absorptive product spectra in the wavenumber window,  $1360\text{ cm}^{-1} \leq \tilde{\nu} \leq 1400\text{ cm}^{-1}$ , and the roots of the polynomial's first derivative were determined numerically. The delay-dependent peak position thus obtained is plotted in Fig. 6a (bottom panel) together with a fit to a single-exponential correlation function

$$\tilde{\nu}(t) = [\tilde{\nu}(0) - \tilde{\nu}(\infty)] \cdot e^{-t/\tau_{\text{VER}}} + \tilde{\nu}(\infty), \quad (11)$$

where the asymptotic band position,  $\tilde{\nu}(\infty)$ , was set to the stationary position of the  $\text{E}(\text{C}-\text{O})$ -band in the FTIR spectrum and only the data for delays up to 100 ps were considered. An optimal fit was obtained with  $\tilde{\nu}(0) = 1363\text{ cm}^{-1}$  together with a ground-state vibrational relaxation time,  $\tau_{\text{VER}}$ , of 2.2 ps. The ultrafast nature of the vibrational relaxation dynamics indicate that they are governed mostly by IVR within the complex itself rather than VET from the

solute to the solvent. The experimental data begin to slightly deviate from the fit for delays in excess of 100 ps. This is likely caused by the finite overlap of the relaxing  $\text{E}(\text{C}-\text{O})$ -band's high-frequency wing with the neighboring induced absorption, which in turn, also experiences some significant spectral shift at longer delays.

A corresponding peak position analysis was therefore also conducted on the purely absorptive product spectra in the frequency region,  $1410\text{ cm}^{-1} \leq \tilde{\nu} \leq 1460\text{ cm}^{-1}$ ; the results of which are displayed in Fig. 6b. It can be seen that during the phase of ground state relaxation, i.e. for delay shorter than 10 ps, this absorption band remains firmly at its initial position of  $1431\text{ cm}^{-1}$ . However, the band begins to slowly creep to higher wavenumbers; in particular, for delays longer than 100 ps. Unfortunately, the maximum pump-probe window of our instrument is too limited to allow for an extraction of a reliable time constant and the asymptotic band position. The solid curve shown in Fig. 6b corresponds to a single-exponential fit (11) using  $\tilde{\nu}(0) = 1331.2\text{ cm}^{-1}$ ,  $\tilde{\nu}(\infty) = 1439.5\text{ cm}^{-1}$ , and a lower limit time constant,  $\tau$ , of 400 ps, and it serves merely as a guide to the eye.



**Fig. 6.** Experimental time-dependent peak positions (symbols) derived from the purely absorptive product spectra in the oxalate  $\text{C}-\text{O}$  region (a and b) and the oxalate  $\text{C}=\text{O}$  region (c). The solid curves represent single (a and b) or multi-exponential (c) fits. The short dashed, long dashed, and dot-dashed curves in panel c) are the three decaying components to the total fit. Vibrational relaxation of the replenished hot ground state of ferrioxalate takes place on a 10-ps time scale.

Finally, a complementary peak position analysis was also carried out in the oxalates' peripheral  $\text{C}=\text{O}$  stretching region between  $1630\text{ cm}^{-1}$  and  $1650\text{ cm}^{-1}$  thereby zooming in on the strong product absorption adjacent to the hot ground state's  $\text{E}(\text{C}=\text{O})$  band. The results are shown in Fig. 6c and turn out to be highly informative. Initially, the band shifts to higher frequencies to reach a maximum peak position of  $1645\text{ cm}^{-1}$  at a time delay of 2.5 ps. Subsequently,

the band maximum shifts continuously back to lower wavenumbers to reach a value of  $1636\text{ cm}^{-1}$  at a delay of 1 ns. Most importantly, the total frequency downshift of  $\sim 11\text{ cm}^{-1}$  occurs in a pronounced triple-exponential fashion. The three individually decaying components as well as the overall multi-exponential fit to the delay-dependent peak position are also shown in Fig. 5c. The three downshifts are characterized by time constants of 2.5 ps, 34 ps, and 400 ps with associated amplitudes of  $2\text{ cm}^{-1}$ ,  $5\text{ cm}^{-1}$ , and  $13\text{ cm}^{-1}$ , respectively. The fastest component of which together with an ultrafast upshift (1.25 ps time constant and  $19\text{ cm}^{-1}$  magnitude) describe the behavior of the data during the first 10 ps, i.e. on a time scale, which is clearly dictated by the vibrational relaxation of the hot replenished ground state and during which the relaxing E(C=O) band still heavily overlaps with the product absorption of interest.

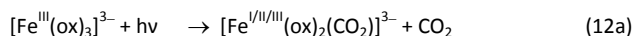
The two slower components with time constants of 34 ps and 400 ps are particularly intriguing as they promptly suggest the existence of a primary fragment (apart from the neutral  $\text{CO}_2$ ) at early times that continues to structurally transform in two consecutive steps. To better characterize the final state, which the system adopts within our limited observation window, one can inspect the purely absorptive product spectrum recorded at the longest delay of 1 ns (cf. Fig. 5b, black curve). From the spectral shift dynamics, it was concluded that the ground state relaxation is essentially complete after 10 ps. Therefore, one can remove the equilibrated ground-state contribution from the 1-ns spectrum by subtracting the weighted stationary FTIR spectrum of the parent using once again the forbidden  $A_1(\text{C=O})$  peak at  $1709\text{ cm}^{-1}$  for selecting the proper weighting factor. The resultant residuals are displayed in Fig. 5b by the blue spectrum. It consists of three distinct peaks located at  $1440\text{ cm}^{-1}$ ,  $1636\text{ cm}^{-1}$ , and  $1665\text{ cm}^{-1}$ , respectively. Their assignment proves critical for an in-depth understanding of the primary processes of ferrioxalate.

## 2.5 Primary photochemical mechanism

The carbon dioxide radical anion,  $\text{CO}_2^{\bullet-}$ , which was discussed above in the context of the possible primary bond breakages following the photon absorption, has been observed before through infrared spectroscopy following Penning ionization of  $\text{CO}_2/\text{Ne}$  mixtures and rapid depositing of the gaseous sample on a cryo-substrate at 5 K.<sup>42</sup> The asymmetric stretching vibration of the radical anion in the rare gas matrix was found to absorb at  $1658\text{ cm}^{-1}$ , i.e. right in the oxalate's C=O stretching region. The  $\text{CO}_2^{\bullet-}$  species could also be stabilized in pressed alkali halide disks, which contained small amounts (1 ppm) of potassium formate, from which it was generated through exposure of the sample to gamma rays from a cobalt source.<sup>43</sup> In this study, the infrared band of the radical anion's asymmetric stretching mode was found at  $1665\text{ cm}^{-1}$ .

Clearly, both of these reports match very nicely the spectral position of the most upshifted product absorption in the oxalate's C=O stretching region after 1 ns and we therefore assign the  $1665\text{ cm}^{-1}$  band in Fig. 5b to the free carbon dioxide radical anion. The quantitative cleavage of a single neutral  $\text{CO}_2$  from each photo-excited parent, which is able to avoid an internal conversion, has already been confirmed above. Moreover, the free oxalate anion,  $\text{ox}^{2-}$ , is known to absorb in aqueous solutions at  $1579\text{ cm}^{-1}$ ,<sup>44</sup> while its

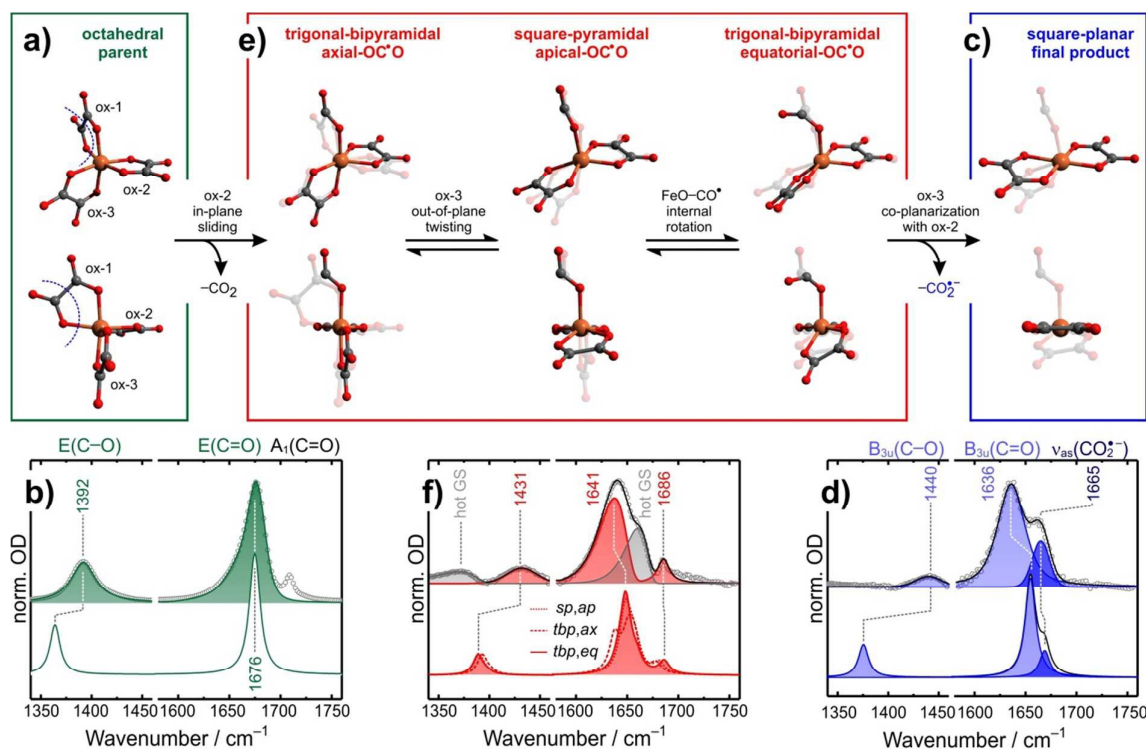
corresponding radical mono-anion,  $\text{C}_2\text{O}_4^{\bullet-}$ , has also been identified in cryogenic argon matrices, where it was found to absorb at  $1857\text{ cm}^{-1}$ .<sup>45</sup> Despite all efforts, we have not been able to detect any appreciable signal in both of these spectral regions for delays of up to 1 ns. Therefore, it seems highly unlikely that within 1 ns, more than one oxalate ligand is cleaved from the iron center. Thus, it must be concluded that by 1 ns, one (and only one) of the three dianionic oxalate ligands has fully fragmented into a singly negatively charged carbon dioxide radical and a neutral  $\text{CO}_2$  molecule. Thus, given that the primary two-body dissociation prevails over the three-body fragmentation, the two consecutive processes



must have occurred within 1 ns and with a quantum yield of 65%. Therefore, we are led to conclude that the two product bands at  $1440\text{ cm}^{-1}$  and  $1636\text{ cm}^{-1}$  arise from the final ferrous dioxalato product. Unfortunately, IR-spectroscopic data on  $[\text{Fe}^{\text{II}}(\text{ox})_2]^{2-}$  are hard to come by because in aqueous solution, this complex coexists with other ferrous species such as  $[\text{Fe}^{\text{II}}(\text{ox})]$  and  $[\text{Fe}^{\text{II}}(\text{ox})_3]^{2-}$ , as well as mixed aquo-oxalato complexes, each of which is readily oxidized in the presence of air to a ferric compound.<sup>46-48</sup> However, IR-spectroscopic data do exist for the dioxalato/trioxalato pair of cobalt complexes,  $[\text{Co}^{\text{III}}(\text{ox})_3]^{3-}$  and  $[\text{Co}^{\text{II}}(\text{ox})_2]^{2-}$ , which feature the same metal oxidation states and symmetries ( $D_{3d}$  and  $D_{2h}$ ) as the Fe(III)-parent/Fe(II)-product pair of the reaction sequence (12).

The oxalate C=O stretching resonance of the Co(III) complex has been reported by Nakamoto and coworkers<sup>31, 49</sup> to appear at  $1670\text{ cm}^{-1}$  whereas that of the Co(II) species was found at  $1625\text{ cm}^{-1}$ . Thus, in the C=O region a downshift of  $45\text{ cm}^{-1}$  is expected upon one-electron reduction and loss of an oxalate ligand. At the same time, C–O absorptions of the Co(III) complex were found at  $1398\text{ cm}^{-1}$  and  $1254\text{ cm}^{-1}$ , whereas those of the Co(II) species were identified at  $1450\text{ cm}^{-1}$  and  $1306\text{ cm}^{-1}$ . Therefore, an upshift of  $52\text{ cm}^{-1}$  is expected for the C–O stretching modes. Transferring these numbers to the ferrioxalate/ferrooxalate system, we would anticipate the C=O resonance of  $[\text{Fe}^{\text{II}}(\text{ox})_2]^{2-}$  to be located at  $1631\text{ cm}^{-1}$  and a C–O band to appear at  $1444\text{ cm}^{-1}$ . These predictions are in perfect accord with our experimental product spectrum recorded at a delay of 1 ns (cf. Fig. 5b). Consequently, the strong induced absorption peaking at  $1636\text{ cm}^{-1}$  is assigned here to the peripheral C=O-stretching mode (mostly species  $B_{3u}$  in point group  $D_{2h}$ ) of a square-planar dioxalatoferrate(II) product complex whereas the  $1440\text{ cm}^{-1}$ -band is attributed to its inner C–O-mode (also  $B_{3u}$ ).

Having spectroscopically identified each the three final products of the reaction (12), the nature of the intermediate species,  $[\text{Fe}^{\text{I/II/III}}(\text{ox})_2(\text{CO}_2)]^{3-}$ , and the origin of the biphasic temporal evolution with time constants of 34 ps and 400 ps remain to be clarified. To this end we have endeavored into a detailed density function theory (DFT) study of a variety of different structures corresponding to the chemical formula,  $[\text{Fe}(\text{ox})_2(\text{CO}_2)]^{3-}$ . For reference, complementary calculations were also carried out on the parent and product complexes,  $[\text{Fe}^{\text{III}}(\text{ox})_3]^{3-}$  and  $[\text{Fe}^{\text{II}}(\text{ox})_2]^{2-}$ , as well as the carbonaceous triatomic fragments,  $\text{CO}_2$  and  $\text{CO}_2^{\bullet-}$ . Geometry optimizations and harmonic normal mode analyses of the resultant equilibrium



**Fig. 7.** Primary photochemical mechanism of ferrioxalate in aqueous solution. a) Structure of the ferric parent from two different perspectives and labeling of the oxalate ligands ox-1 through ox-3. A  $\text{CO}_2$ -molecule is released from ox-1 following absorption of the photon. b) Comparison between the experimental (symbols) and the theoretical IR spectrum (bottom curve) of ferrioxalate. c) Structure of the ferrous parent. d) Experimental (symbols) product spectrum recorded at a delay of 1 ns and its decomposition into two individual components representing the ferrous product (light blue) and the  $\text{CO}_2^{\bullet-}$  radical anion (dark blue). The corresponding theoretical spectrum and its individual product components are shown at the bottom. e) Structure of the isomers of the ferrous dioxalato intermediate carrying a  $\text{CO}_2^{\bullet-}$  radical anion ligand. The primary product is most likely the *tbp*-[Fe(ox)<sub>2</sub>(ax-OCO)]<sup>3-</sup> isomer (left), which can structurally equilibrate with the energetically nearby *sap*-[Fe(ox)<sub>2</sub>(ap-OCO)]<sup>3-</sup> (middle) and *tbp*-[Fe(ox)<sub>2</sub>(eq-OCO)]<sup>3-</sup> (right) isomers. f) Experimental (symbols) product spectrum recorded at a delay of 500 fs and its decomposition into two individual components representing the ferrous intermediate (red) and the hot ground state of the parent (gray). The theoretical spectra of the three ferrous intermediates are shown at the bottom. For the spectral decompositions in panels b), d), and f), Lorentzian, Gaussian, and/or log-normal line shapes were used.

structures were carried out on the doublet, quartet, and sextet surfaces for all triply anionic Fe-species and on the singlet, triplet, and quintet surfaces for the Fe-containing dianionic product. The results of such calculations at the level of theory, B3LYP<sup>50-51</sup>/def2-TZVP<sup>52</sup>/PCM(H<sub>2</sub>O)<sup>53</sup>, are compiled in Fig. 7.

Even without empirically scaling the harmonic frequencies,<sup>54</sup> the method is able to reproduce the experimental FTIR spectrum of the parent (panels a and b). Note that the minor  $A_1(\text{C}=\text{O})$  cannot not show up in the theoretical spectrum because the parent's optimized structure is perfectly  $D_3$ -symmetric. Moreover, the same method predicts a composite theoretical IR spectrum for the final products of reaction (12), which is also in excellent agreement with the experimental product spectrum obtained at 1 ns (cf. panels c and d). In agreement with the experimental observation, the  $B_{3u}(\text{C}=\text{O})$ -band of the dioxalatoferat(II) product is shifted to lower wavenumbers as compared the  $E(\text{C}=\text{O})$ -band of the parent while the corresponding  $\text{C}-\text{O}$  band is shifted to higher wavenumber relative to that of the parent. In addition, the asymmetric stretching absorption of the carbon dioxide radical anion finds itself on the

high-frequency edge of the  $B_{3u}(\text{C}=\text{O})$ -band, again, in perfect harmony with the experimental findings.

The consensus between experiment and theory for both, the ferric reactant and the final products of the reaction (12), lends significant credence to spectroscopic predictions for possible intermediates at the particular level of theory chosen. Indeed, the calculations identify a number of isomeric complexes of the type, [Fe(ox)<sub>2</sub>(CO<sub>2</sub>)]<sup>3-</sup>, which differ with regard to the mode of binding between the iron center and the lone CO<sub>2</sub>-ligand. A full account on these isomers has recently been published elsewhere.<sup>55</sup> On the low-spin surface ( $S = 1/2$ ), the most stable isomer exhibits a  $\eta^2_{\text{C},\text{O}}$ -binding mode, in which a bent CO<sub>2</sub>-moiety is "side-on" coordinated to the metal and the spin density resides exclusively on the Fe-center. Formally, this species can be regarded as an iron(III) complex that features a quite unique ferraoxacyclopropanone unit. However, its IR-spectrum is incompatible with the experimental spectrum recorded at a delay of 500 fs and it must therefore be dismissed as the primary Fe-containing fragment.

On the high-spin surface ( $S = 5/2$ ), the most stable isomer displays a  $\eta^1_{\text{C}}$ -binding mode, in which a bent CO<sub>2</sub>-moiety is coordinated

to the metal through the central carbon atom. Significant spin density resides on the CO<sub>2</sub>-ligand such that the metal can be regarded as a ferrous, i.e. Fe(II), center. The IR-spectrum of this species is also found to be in stark contrast to the 500-fs product spectrum, which is why this species can neither be the primary fragment.

However, on the sextet surface, three additional isomers were found that are energetically very close to, but structurally markedly different from the above  $\eta^1_C$ -complex. Each of these three species belongs to a  $\eta^1_O$ -binding motif, in which a bent OCO-moiety is attached to the metal center through one of the two terminal O-atoms. Notably, spin density equivalent to a full unpaired electron is delocalized over OCO-ligand. The electronic structure of these species is therefore best described by a carbon dioxide radical anion ligand,  $-(O-C-O)^{\bullet-}$  ( $S = \frac{1}{2}$ ), that is ferromagnetically coupled to a high-spin ( $S = 2$ ) iron(II) center, which in turn is bearing two dianionic oxalate ligands, thereby giving rise to complexes of the type  $[Fe^{II}(ox)_2(-OCO)]^{3-}$ . In detail, the three complexes differ with respect to the geometry of the ligand sphere as depicted in Fig. 7e. One isomer exhibits a slightly distorted square-pyramidal (*sp*) coordination pattern with the OCO-ligand occupying the apical (*ap*) binding site whereas the other two isomers feature a trigonal-bipyramidal (*tbp*) coordination sphere with the OCO-ligand residing either in the equatorial (*eq*) or the axial (*ax*) position. The *sp*- $[Fe(ox)_2(ap-OCO)]^{3-}$  isomer is 5.3 kJ/mol (or equivalently, 442 cm<sup>-1</sup>) and 6.5 kJ/mol (544 cm<sup>-1</sup>) more stable than the *tbp*- $[Fe(ox)_2(ax-OCO)]^{3-}$  and *tbp*- $[Fe(ox)_2(eq-OCO)]^{3-}$  isomers, respectively.

Most importantly, the theoretically predicted IR-spectra of all three isomers are in remarkable agreement with the purely absorptive product spectrum recorded at a delay of 500 fs (cf. Fig. 7f). Each of the three complexes displays a very strong absorption due to the peripheral C=O stretching vibrations of the intact two oxalate ligands, which is markedly downshifted with respect to the equivalent E(C=O) band of the ferric parent and simultaneously, visibly upshifted relative to the B<sub>3u</sub>(C=O)-band of the final ferrous dianionic product. At the same time, the three species display a weaker absorption arising from the inner C–O stretching modes, which is clearly upshifted relative to the corresponding E(C–O) band of the parent. Finally, all three isomers are characterized by a minor peak due to the asymmetric stretching vibration of the (–OCO)-radical anion ligand that is significantly upshifted with regard to the dominant band of the peripheral C=O modes. Taken together, these findings constitute very compelling evidence that the primary Fe-containing fragment, which is responsible for the 500 fs-spectrum shown in Fig. 5a, is indeed a ferrous complex of the type  $[Fe^{II}(ox)_2(-OCO)]^{3-}$  featuring the  $\eta^1_O$ -mode of CO<sub>2</sub>-binding.

An inspection of the molecular structures reveals that only a minimal amplitude motion is required to transform the  $[Fe^{III}(ox)_3]^{3-}$  parent complex into the *tbp*- $[Fe^{II}(ox)_2(ax-OCO)]^{3-}$  isomer. Following the Fe–O and C–C bond cleavages and removal of the light neutral CO<sub>2</sub> fragment from one of the three equivalent oxalate ligands (ox-1 in Fig. 7), only a subtle 30°-in-plane sliding motion of either of the two remaining intact oxalates (here, ox-2) is required to convert the distorted octahedral ligand field of the parent into the trigonal-bipyramidal coordination sphere of the primary product. Because of

its sliding nature, this motion cannot be hindered much by solvent friction and may therefore very well occur on a time scale of 500 fs.

To convert the structure of *tbp*- $[Fe^{II}(ox)_2(ax-OCO)]^{3-}$  to the slightly more stable isomer, *sp*- $[Fe^{II}(ox)_2(ap-OCO)]^{3-}$ , an out-of-plane twisting of the third oxalate ligand (ox-3 in Fig. 7) is required. This internal rotatory motion of ox-3 is of large amplitude and will thus experience a much stronger frictional damping as compared to the initial small amplitude sliding motion of ox-2. Furthermore, *tbp*- $[Fe^{II}(ox)_2(ax-OCO)]^{3-}$  is nearly isoenergetic with *tbp*- $[Fe^{II}(ox)_2(eq-OCO)]^{3-}$  but their interconversion requires also some out-of-plane twisting of ox-3 in addition to torsional motion about the FeO–CO dihedral associated with the OCO-radical anion ligand. Therefore, we can expect the primary Fe-containing fragment to equilibrate with its two structural isomers, but these dynamics will occur on a much longer time scale due to solvent friction. The dominant induced absorption in the oxalate's C=O stretching region was seen to experience a pronounced frequency downshift on the time scale between 10 ps and 100 ps (cf. Fig. 6c). A time constant for this spectral shift of 34 ps was determined. Here, we attribute this spectro-temporal evolution and the associated time constant to the structural equilibration of the primary ferrous photo-fragment with its isomeric forms as depicted in Fig. 7e. Remembering that the final product spectrum at a delay of 1 ns could be assigned unequivocally to the final product species of the actinometer reaction (1), namely  $[Fe^{II}(ox)_2]^{2-}$ , free CO<sub>2</sub><sup>•-</sup>, and free CO<sub>2</sub>, it must be concluded that the final phase of the spectral downshift in Fig. 6c reflects the dissociation dynamics of the carbon dioxide radical anion from the equilibrium mixture of penta-coordinated primary ferrous fragments.

However, the question then arises as to why the final bond breakage with loss of CO<sub>2</sub><sup>•-</sup> is so slow. Note, it is an order of magnitude slower than the preceding equilibration between the isomeric primary products. To solve this puzzle one can resort again to the DFT results and examine specifically the energetics of the step (12b). It turns out that an infinitely separated dioxalatoferate(II) dianion together with a carbonaceous radical anion is actually much higher in energy than all three penta-coordinated precursors. The corresponding dissociation energy,  $E_D$ , amounts to 31.9 kJ/mol or equivalently, 2669 cm<sup>-1</sup>, for the least stable of the three isomers, *tbp*- $[Fe^{II}(ox)_2(eq-OCO)]^{3-}$ . It was already noticed above that quite likely the neutral carbon dioxide fragment, which is expelled during the first step of sequence (12), remains in close proximity to the penta-coordinated intermediate. The two carbonaceous fragments may therefore combine to a loose dimer complex



which has indeed been observed in previous theoretical<sup>56</sup> and experimental studies.<sup>57</sup> The DFT-calculations predict this process to release an energy of 3.4 kJ/mol (i.e. 286 cm<sup>-1</sup>) thereby possibly lowering the dissociation energy to 28.5 kJ/mol (or 2383 cm<sup>-1</sup>). At any rate, the second step in sequence (12) is clearly an uphill process, thereby rationalizing why it occurs on a much longer time scale. According to the theory, the Fe–OCO bond stretching vibration has a frequency of roughly 230 cm<sup>-1</sup>, which can be used as an attempt frequency,  $\nu_a$ , for estimating a dissociation rate constant,  $k_D$ . Using these numbers together with

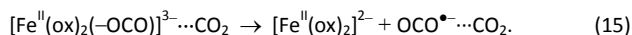
$$k_D = \nu_D \exp(-E_D/k_B T) \quad (14)$$



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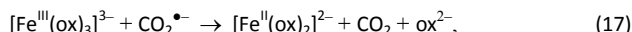
we arrive at a time constant of  $1/k_D = 2.6$  ns when a temperature  $T = T(\infty) = 350$  K from above is used. Compared to the experimental time constant, this value is somewhat too large but it nonetheless demonstrates that the formation of the final products from the penta-coordinated intermediate is clearly retarded even if it occurs as an in-cage ligand abstraction by the nearby  $\text{CO}_2$



Hence, we lastly assign the time constant of 400 ps to the Fe–O bond breakage between intermediate's Fe-center and the  $\text{CO}_2^{\bullet-}$  ligand; a thermal dissociation, which is necessary for the formation of the final dioxalatoferat(II) complex. Finally, it should be pointed out that the loose carbonaceous complex may isomerize to the oxalate radical anion according to



Our theoretical calculations predict an energy release of 14.2 kJ/mol (or  $1185 \text{ cm}^{-1}$ ) for this process; however, as mentioned above, a spectroscopic signature of the  $\text{C}_2\text{O}_4^{\bullet-}$  species could not be recorded for delays of up to 1 ns. At any rate, all open-shell fragments,  $\text{CO}_2^{\bullet-}$ ,  $\text{C}_2\text{O}_4^{\bullet-}$ , and  $\text{OCO}^{\bullet-} \cdots \text{CO}_2$  are highly reactive species that can easily attack another ferric parent in the bulk



thereby explaining an overall quantum yield for the ferrioxalate actinometer that is larger than one.

### 3 Conclusion

In summary, we have measured the primary events of the ferrioxalate actinometer using UV-pump/MIR-probe spectroscopy. The photochemical dynamics occurring within the first nanosecond after 266 nm excitation can be summarized as follows:

Following absorption of the 266 nm photon, the electronically excited state of the ferrioxalate trianion dissociates within 500 fs into neutral, vibrationally hot carbon dioxide and a penta-coordinated ferrous dioxalate intermediate complex that carries a bent carbon dioxide radical anion ligand. The initial Fe–O and C–C bond breakages compete with internal conversion, which limits the primary quantum yield for the initial photoreduction to roughly 69%. The neutral  $\text{CO}_2$ -fragment is seen to vibrationally relax with time constants of 1.25 ps and 25 ps in a mode-specific fashion and the fragment recoil is likely to be responsible for the fast cooling of the bending and symmetric stretching vibrations of the triatomic primary product. Simultaneously, the replenished parent ground state undergoes intramolecular vibrational relaxation with a time constant of 2.2 ps. The primary ferrous product exists in three different isomeric forms, which can structurally interconvert on a time scale of 40 ps. On a nanosecond time scale, the reactive dynamics are completed by a removal of the  $\text{CO}_2^{\bullet-}$  ligand from the coordination sphere surrounding the ferrous center, thereby yielding the dioxalatoferate(II), a free  $\text{CO}_2^{\bullet-}$  together with a neutral free  $\text{CO}_2$  molecule. The latter two species may form a loose complex on a time scale of 1 ns, which may assist in lowering the reaction barrier for the decay of the primary ferrous intermediate.

The primary yield for reaction (7) of 69% implies a total photochemical quantum yield for iron(II) formation of  $2 \times 0.69 = 1.38$  if

every  $\text{CO}_2^{\bullet-}$  radical anion were able to react in the bulk with another ferrioxalate parent complex in a secondary bimolecular reaction such as (17). However, this value is slightly higher than the experimental quantum yield of the actinometer of 1.25 at 266 nm suggesting that a minor fraction of 18% of the carbonaceous fragments is lost through competing reaction channels such as hydrogen abstraction from the solvent to produce formate or diffusive self-recombination to free oxalate anions,  $\text{C}_2\text{O}_4^{2-}$ . We will clarify the fate of the radical anion in future step-scan Fourier-transform infrared studies on nanosecond time scales.

The primary mechanism that our experimental data suggests falls into the category of "prompt" electron transfer. The release of the neutral  $\text{CO}_2$  fragment occurs in less than 1 ps and the metal-containing species is clearly already a ferrous species. The electron transfer from the ligand sphere to the metal may already be caused by the photon absorption if we regard the electronic excitation as a full one-electron ligand-to-metal charge transfer transition. At the latest, the Fe–O and C–C bond breakages that are required for the  $\text{CO}_2$  departure from the ligand sphere will bring about or complete the full one-electron reduction at the iron center.

Finally, we emphasize that the penta-coordinated ferrous intermediate represents a highly intriguing class of carbon dioxide complexes, which have not been seen before.<sup>55</sup> The species revealed here on the femtosecond to picosecond time scale carries the reductively activated form of the inert greenhouse gas,  $\text{CO}_2$ ; namely, the carbon dioxide radical anion. As such, the intermediate promises highly intriguing chemical reactivity, which we are currently exploring in more detail in our laboratories.

### Conflicts of interest

There are no conflicts to declare.

### Acknowledgements

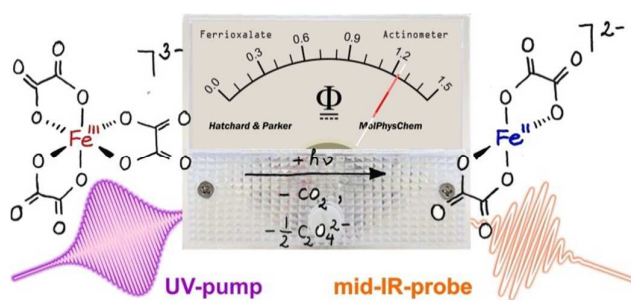
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## Table of Contents graphic



Ultrafast vibrational spectroscopy unravels the primary processes of Hatchard and Parker's ferrioxalate actinometer – a widely used tool in the photochemistry community.